

Metro Ticket Booking System

Comprehensive Project Execution Plan (Planning Phase)

PLATFORM TARGET
ServiceNow App Engine

APPLICATION SCOPE
Scoped App (x_snc_metro_book)

DOCUMENT VERSION
v1.0 Final

1. Work Breakdown Structure (WBS) & Deliverables

- 1. Initiation & Scoping (Weeks 1-2)
 - 1.1 Scope Definition & Architecture Sign-off
 - 1.2 Development Instance Setup & Scope Provisioning (x_snc_metro_book)
- 2. Core Data & Platform Setup (Weeks 3-4)
 - 2.1 Table Schema Configuration (sn_metro_station, sn_metro_ticket, etc.)
 - 2.2 Security Architecture (ACLs & Access Roles)
 - 2.3 Decision Tables Setup (Peak Hours & Fare Matrix Rules)
- 3. Service Logic & Core Development (Weeks 5-7)
 - 3.1 Script Includes Development (HMAC/TOTP QR Generator & Fare Calculators)
 - 3.2 Inbound Scripted REST APIs (Sub-150ms Turnstile Validation Endpoints)
 - 3.3 Flow Designer Automation (Post-Payment & Auto-Refund Orchestration)
- 4. Portal & Interface Experience (Weeks 8-9)
 - 4.1 Service Portal Custom Widgets (Station Selectors & Live QR Canvas)
 - 4.2 Mobile App Setup (Now Mobile Offline QR Storage)
 - 4.3 Operational Workspace (UI Builder Workspace for Station Masters)
- 5. Integrations & Testing (Weeks 10-11)
 - 5.1 Payment Gateway Integration Hub Spokes (Stripe/Razorpay Integration)
 - 5.2 Turnstile Hardware Mock Testing (Load Testing <200ms Response Times)
 - 5.3 Automated Testing Framework (ATF Test Suites)
- 6. Deployment & Hypercare (Week 12+)
 - 6.1 Update Set / App Repo Packaging & Deployment
 - 6.2 Production Cutover & Hypercare Monitoring

2. Milestone Schedule & Timeline

Phase	Milestone Description	Duration	Target Completion	Key Output
Phase 1	Requirement Sign-off & Architecture Baseline	2 Weeks	End of Week 2	Approved FSD
Phase 2	Schema & Core Platform Build	2 Weeks	End of Week 4	Custom Tables, ACLs, Decision Tables
Phase 3	Core Application Logic & Integration Endpoints	3 Weeks	End of Week 7	Gate Validation API & Fare Engines
Phase 4	Commuter UI & Mobile Experiences	2 Weeks	End of Week 9	Service Portal & Offline Mobile App
Phase 5	Integration, Testing & Security Audit	2 Weeks	End of Week 11	ATF Test Pass Report & Performance Metrics
Phase 6	Go-Live & Hypercare Support	1 Week+	Week 12	App Repo Production Deployment

3. Resource & RACI Matrix

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

Project Activity	System Architect	ServiceNow Dev	Integration Lead	Station Mgr	Sponsor
Data Schema & Table Extensions	A	R	C	I	I
Fare Logic & Decision Tables	A	R	I	C	C
Turnstile Scripted REST API	A	R	C	I	I
Payment Hub & Webhooks	A	C	R	I	I
Service Portal & Dynamic QR Canvas	A	R	I	C	I
ATF & Stress Testing (<200ms)	A	R	C	I	I
Production Deployment Sign-off	A	I	I	C	A

4. Key Project Risks & Mitigation Plan

Risk Description	Severity / Prob.	Impact Area	Mitigation Strategy
Turnstile Latency Spikes (>200ms)	High / Medium	Gate Congestion	Indexed composite keys on sn_metro_ticket; offload post-scan updates to Async Business Rules.
No Cellular Network Underground	High / High	QR Render Failure	Enable offline token caching on Now Mobile App so QR codes generate locally without cell signal.
Screenshot Fraud & Ticket Reuse	Medium / High	Revenue Leakage	Implement TOTP-based dynamic QR refresh (30s rotation) rendering static screenshots useless.
Payment Gateway Callback Failure	Medium / Medium	Financial Discrepancy	Deploy a self-healing 60s Flow Designer timeout engine that auto-initiates payment refunds.

5. Definition of Done (DoD) Checklist

[✓] Code Quality & Standards: All Script Includes and REST endpoints pass ServiceNow Scoped Code Review guidelines.
[✓] Performance SLA Compliance: Turnstile Inbound REST API (/api/x_metro/gate_validation) responds in <150ms under peak load.
[✓] Security & Access Control: Strict scoped application roles (x_snc_metro_book.user, x_snc_metro_book.admin) enforced via ACLs.
[✓] Automated Testing: 100% pass rate across Automated Test Framework (ATF) test suites for core purchase and gate flows.
[✓] Offline Capability: Verified local QR token rendering on mobile devices with zero network connectivity.